

PyCharm IDE interface showing a Python script named `project2.py` and its execution output.

Project Structure:

- PyCharmMiscProject
 - .venv library root
 - energy_usage.csv
 - expenses.csv
 - list.py
 - project1.py
 - project2.py
 - External Libraries
 - Scratches and Consoles

Code in `project2.py`:

```
1 import pandas as pd
2 import numpy as np
3
4 # Load CSV
5 df = pd.read_csv("energy_usage.csv")
6
7 # Convert timestamp
8 df['timestamp'] = pd.to_datetime(df['timestamp'])
9
10 # Clean energy column
11 df['energy_kwh'] = df['energy_kwh'].astype(float)
12
13 # Show cleaned dataset
14 print("Cleaned Dataset:")
15 print(df)
16
17 # Total and average energy
```

Run Output:

```
C:\Users\acer\PyCharmMiscProject\.venv\Scripts\python.exe C:\Users\acer\PyCharmMiscProject\project2.py
Cleaned Dataset:
  device_id  room_id  timestamp  energy_kwh
0         1      101 2026-05-13 10:00:00      2.5
1         2      102 2026-05-13 10:10:00      5.2
2         3      103 2026-05-13 10:20:00      1.1
3         1      101 2026-05-13 11:00:00      3.0
```

PyCharm IDE interface showing the continuation of the Python script `project2.py` and its execution output.

Code in `project2.py`:

```
19 average_energy = np.mean(df['energy_kwh'])
20
21 print("\nTotal Energy Used:", total_energy)
22 print("Average Energy Used:", average_energy)
23
24 # Device-wise summary
25 device_summary = df.groupby('device_id')['energy_kwh'].sum()
26
27 print("\nEnergy per Device:")
28 print(device_summary)
29
30 # Room-wise summary
31 room_summary = df.groupby('room_id')['energy_kwh'].sum()
32
33 print("\nRoom-wise Energy Summary:")
34 print(room_summary)
```

Run Output:

```
Total Energy Used: 18.6
Average Energy Used: 3.1

Energy per Device:
device_id
1      5.5
```

PyCharmMiscProject > project2.py 34:20 CRLF UTF-8 4 spaces Python 3.14 (PyCharmMiscProject)

PyCharm IDE interface showing a Python project named "PyCharmMiscProject". The project structure includes a ".venv" library root, "energy_usage.csv", "expenses.csv", "list.py", "project1.py", and "project2.py". The "project2.py" file is open, displaying the following code:

```
19 average_energy = np.mean(df['energy_kwh'])
20
21 print("\nTotal Energy Used:", total_energy)
22 print("Average Energy Used:", average_energy)
23
24 # Device-wise summary
25 device_summary = df.groupby('device_id')['energy_kwh'].sum()
26
27 print("\nEnergy per Device:")
28 print(device_summary)
29
30 # Room-wise summary
31 room_summary = df.groupby('room_id')['energy_kwh'].sum()
32
33 print("\nRoom-wise Energy Summary:")
34 print(room_summary)
```

The "Run" output shows the following results:

```
Name: energy_kwh, dtype: float64
Room-wise Energy Summary:
room_id
101    5.5
102   10.0
103    3.1
```

The status bar at the bottom indicates the file encoding is UTF-8, 4 spaces, and Python 3.14 (PyCharmMiscProject).